# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 8 — MATHS · HARD LEVEL UNIT TEST SET 2

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | RBSE Pattern | Session 2026–27

**Time: 2 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 15 × 1 = 15

*Choose the correct option and write it in your answer book.*

**1.** The difference of the squares of two consecutive natural numbers is 59. The numbers are: 1

**(a)** 29 and 30 **(b)** 29 and 31 **(c)** 30 and 31 **(d)** 28 and 29

**2.** The ones digit of $7^{23}$ is: 1

**(a)** 5 **(b)** 7 **(c)** 3 **(d)** 9

**3.** The number of 1s in the binary form of 77 is: 1

**(a)** 5 **(b)** 4 **(c)** 3 **(d)** 6

**4.** The diagonals of a rectangle ABCD meet at O. If $\angle AOD = 100^\circ$, then $\angle OAB$ is: 1

**(a)** $65^\circ$ **(b)** $50^\circ$ **(c)** $100^\circ$ **(d)** $40^\circ$

**5.** The smallest 3-digit palindrome divisible by 3 is: 1

**(a)** 121 **(b)** 111 **(c)** 101 **(d)** 114

**6.** The number of digits in the square root of a 5-digit perfect square is: 1

**(a)** 3 **(b)** 2 **(c)** 5 **(d)** 4

**7.** If $3^x = 243$, then $3^{x-2}$ equals: 1

**(a)** 2187 **(b)** 241 **(c)** 9 **(d)** 27

**8.** The Roman numeral for 1994 is: 1

**(a)** MCMXCIII **(b)** MCMXCIV **(c)** MMIV **(d)** MCMXCV

**9.** In an isosceles trapezium ABCD with AB $\parallel$ DC, $\angle A = (3x + 10)^\circ$ and $\angle D = (4x + 30)^\circ$. The measure of $\angle C$ is: 1

**(a)** $110^\circ$ **(b)** $145^\circ$ **(c)** $70^\circ$ **(d)** $130^\circ$

**10.** If the number 6a2b is divisible by 90, then $a + b$ equals: 1

**(a)** 2 **(b)** 1 **(c)** 4 **(d)** 3

**11.** $\sqrt[3]{-729}$ is: 1

**(a)** -9 **(b)** 9 **(c)** -10 **(d)** 10

**12.** The value of $(2^{-1} + 3^{-1})^{-1}$ is: 1

**(a)** $\frac{1}{5}$ **(b)** $\frac{6}{5}$ **(c)** 6 **(d)** $\frac{5}{6}$

**13.** **Assertion (A):** The adjacent angles of a parallelogram are supplementary. 1
**Reason (R):** The opposite sides of a parallelogram are parallel, so adjacent angles are co-interior angles on a transversal.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**14.** **Assertion (A):** The number 4,236 is divisible by 3. 1
**Reason (R):** A number is divisible by 3 if the sum of its digits is divisible by 3.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**15.** **Assertion (A):** The square of an even number is always even. 1
**Reason (R):** The sum of the first $n$ odd natural numbers is $n^2$.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

## SECTION — B (Very Short Answer Type) — 5 × 2 = 10

*Answer in about 20–30 words each.*

**16.** The difference between the squares of two consecutive numbers is 75. Find the numbers. 2

**17.** Write $(3 \times 10^5) + (4 \times 10^4)$ in scientific notation. 2

**18.** Write 90 in binary and count the number of 1s in it. 2

**19.** The angles of a quadrilateral are in the ratio 2:5:6:7. Find the largest angle and say whether it is acute, right or obtuse. 2

**20.** How many multiples of 11 lie between 150 and 800 (both included)? 2

## SECTION — C (Short Answer Type) — 5 × 3 = 15

*Answer in about 40–60 words each.*

**21.** Add $343_5$ and $212_5$ (numbers in base 5) and give the answer in base 5. 3

**22.** The angles of a quadrilateral are $x^\circ$, $(2x - 10)^\circ$, $(3x - 20)^\circ$ and $(4x - 30)^\circ$. Find all four angles. 3

**23.** Find the greatest 4-digit number divisible by 6, 8 and 12. 3

**24.** A rectangular hall 36 m by 48 m is to be paved with equal square tiles, the largest possible. Find the side of each tile and the number of tiles. 3

OR

Find the value of $\dfrac{2^6 + 2^5}{2^4 \times (2 + 1)}$. 3

**25.** How many different numbers can be written using at most 8 digits in base 2? What is the largest of them in decimal? 3

OR

One angle of a parallelogram is 2 times its adjacent angle. Find all the angles of the parallelogram. 3

## SECTION — D (Long Answer Type) 2 × 5 = 10

*Answer in about 100–150 words each.*

**26.** **Case study.** The distance of the Earth from the Sun is about $150\,000\,000$ km and light travels at about $3 \times 10^5$ km per second. Students of a science club use these to answer questions on standard form. 5

(i) Write $150\,000\,000$ in standard form.

(ii) Find the time light takes to reach the Earth from the Sun.

(iii) Write that time in standard form.

(iv) Light takes about 1.3 s to reach the Earth from the Moon. Find the Moon's distance in standard form.

**27.** Body A has mass $6 \times 10^{24}$ kg and body B has mass $3 \times 10^{22}$ kg. Which is heavier? How many times as heavy is it? Find the total mass in scientific notation. 5

OR

Subtract $1222_5 - 341_5$ in base 5. Verify your answer in decimal. 5

**END OF QUESTION PAPER** Total: 50 Marks

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